The Bank of England Speech Analysis

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<u>Team members:</u>

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Background and Objective

Our objective is to analyse sentiment in Bank of England speeches to address strategic questions on economic impact and potential predictive insights.

Key areas of focus:

- Changes in speech sentiment over time
- Correlation between sentiment and major economic events
- Links between sentiment and economic indicators
- The predictive power of speeches

Methodology

Retrieve & Clean Data

1. Download all relevant data
2. Clean in Excel and Python

Exploratory Data Analysis

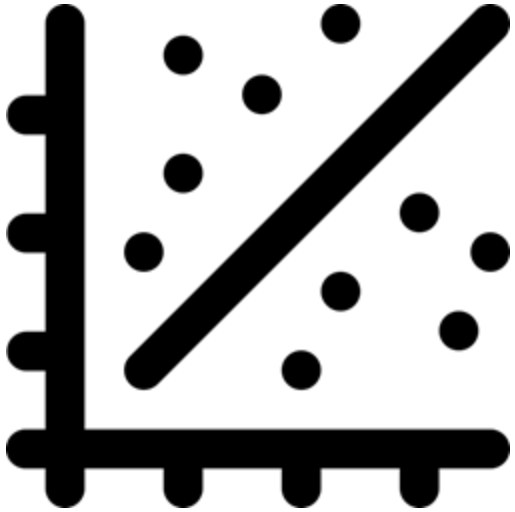
3. Identified key themes
4. Examined data distributions and trends

Sentiment Analysis

5. Modelled speech sentiment using FinBERT
6. Compared with Loughran-Mcdonald dictionary

Analyse Correlation with Economic Indicators

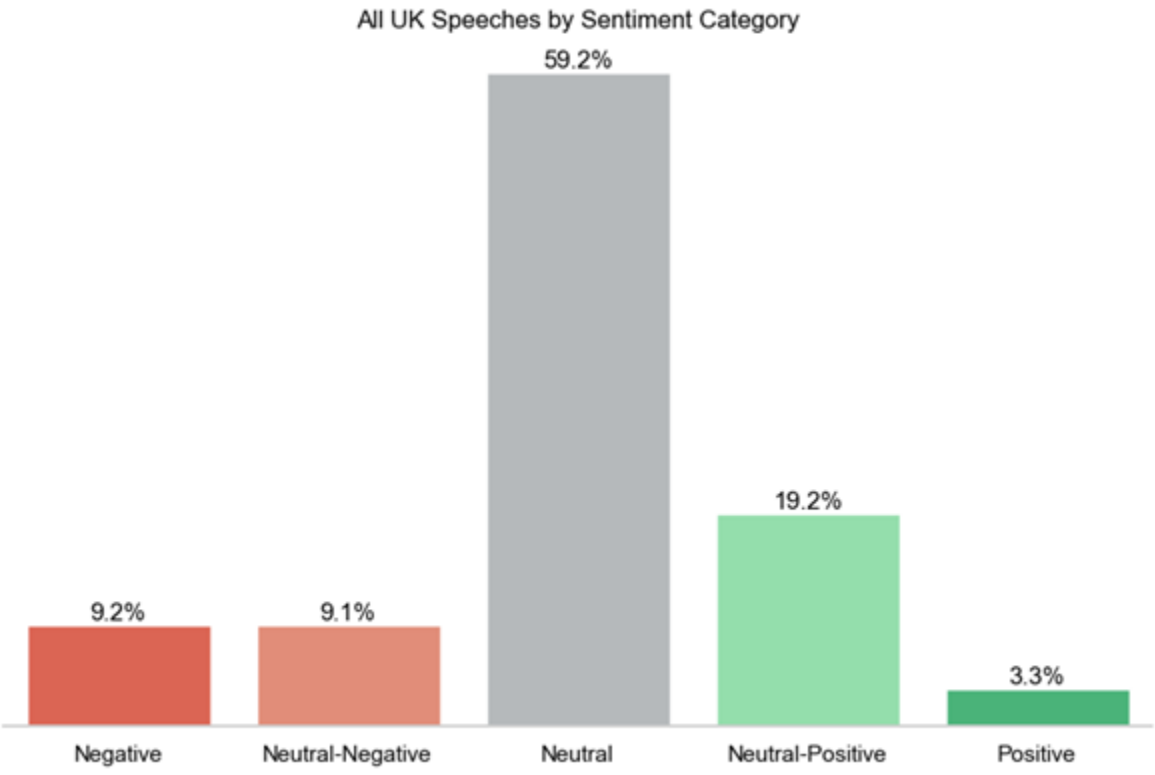
7. Developed regression models for predictive analytics

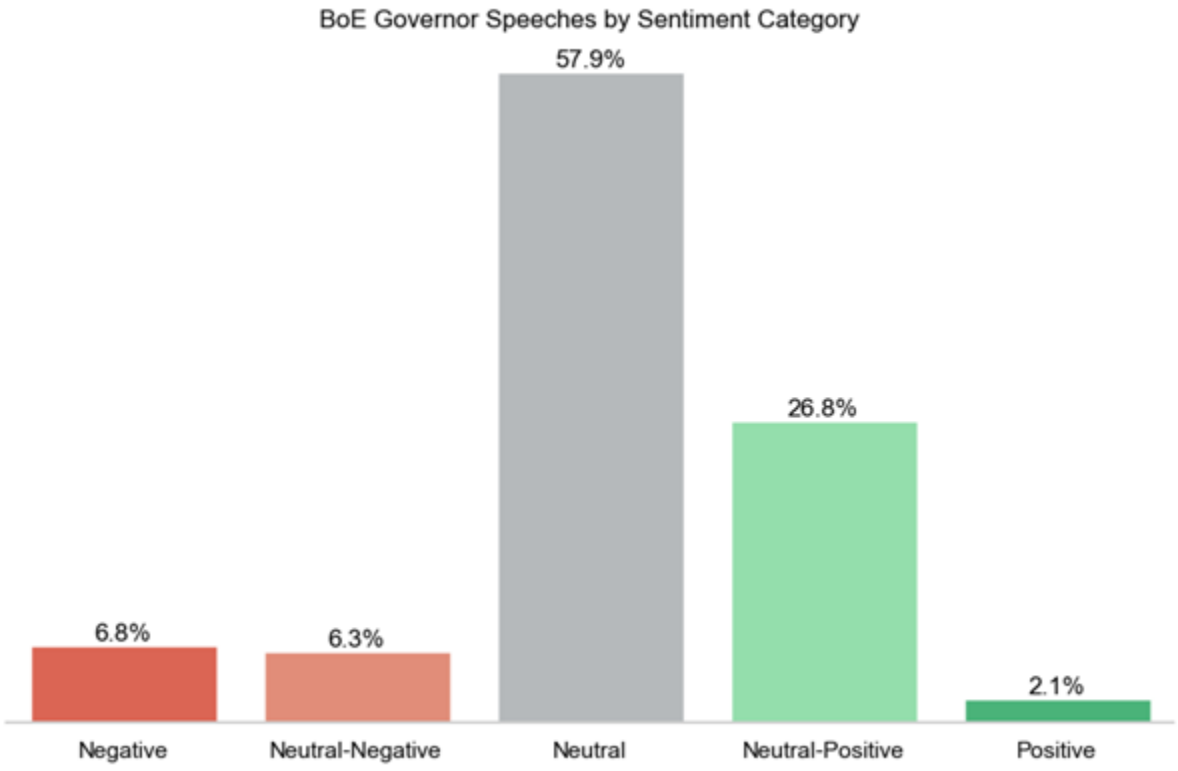


FinBERT Speech Classification

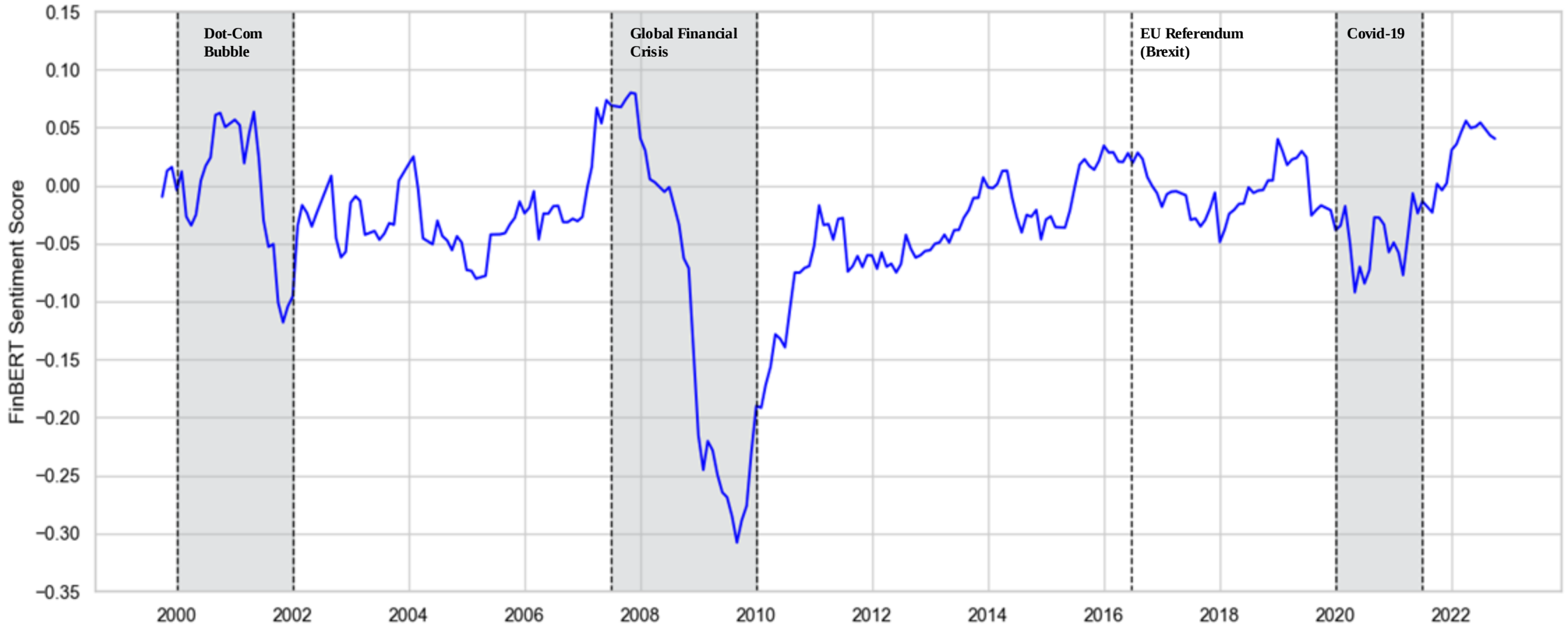
Adjusted FinBERT classification to include ‘Neutral-Negative’ and ‘Neutral-Positive’.

- BoE Governor speeches have a lower frequency of extreme sentiment (Either ‘Positive’ or ‘Negative’).
- Speeches by a BoE Governor have a stronger tendency of being classified ‘Neutral-Positive’.



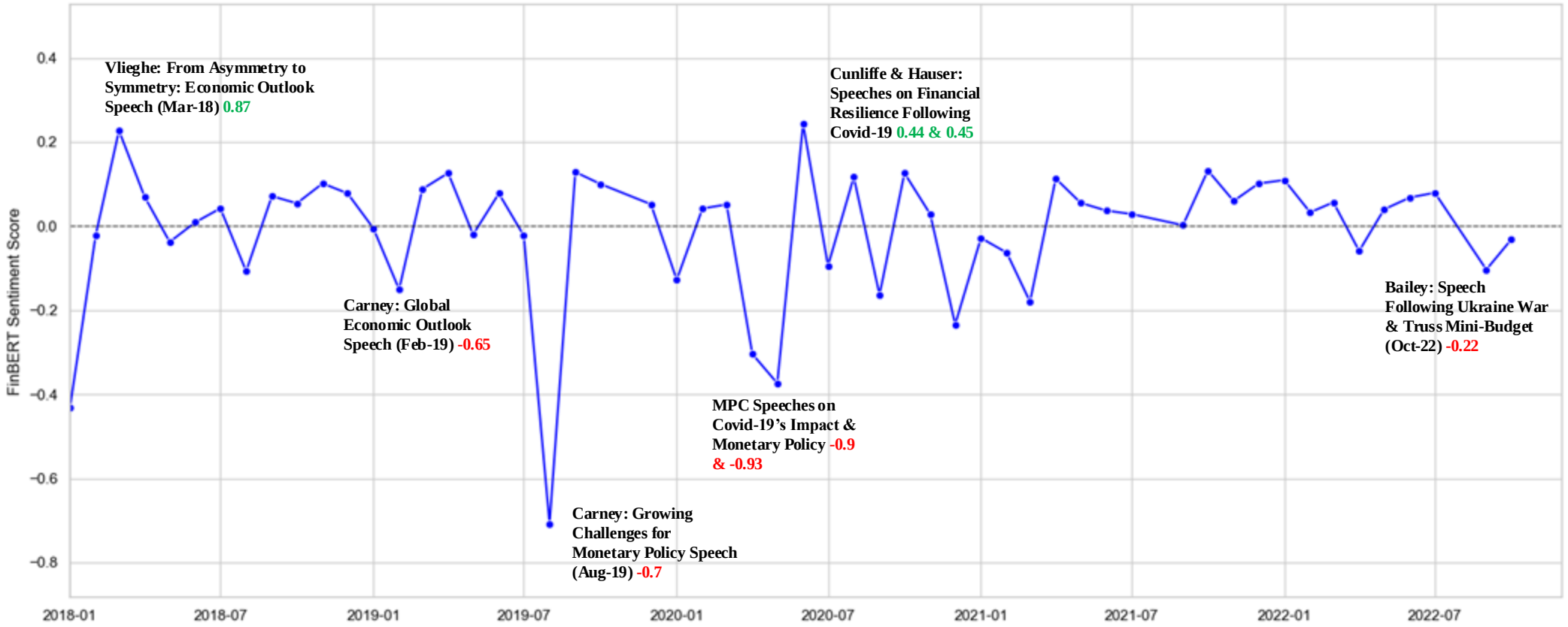


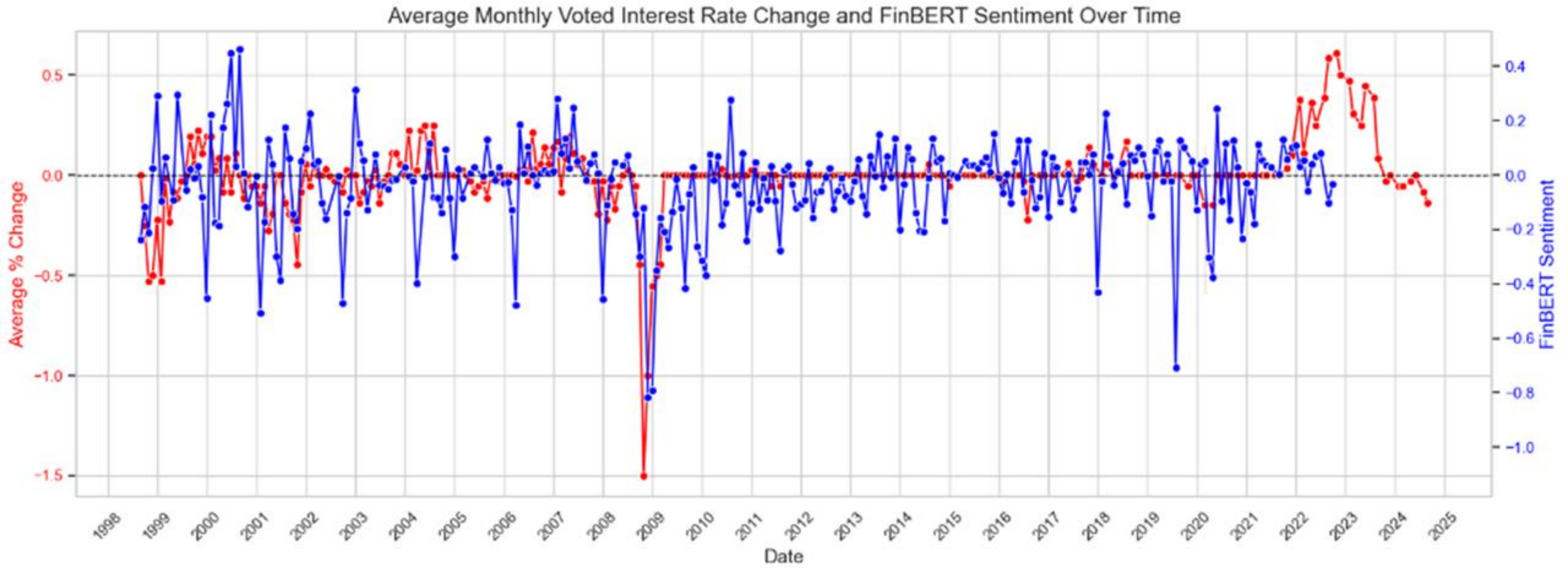
UK Speech Sentiment: 12-Month Moving Average



**Sentiment Score: P(Positive) – P(Negative)*

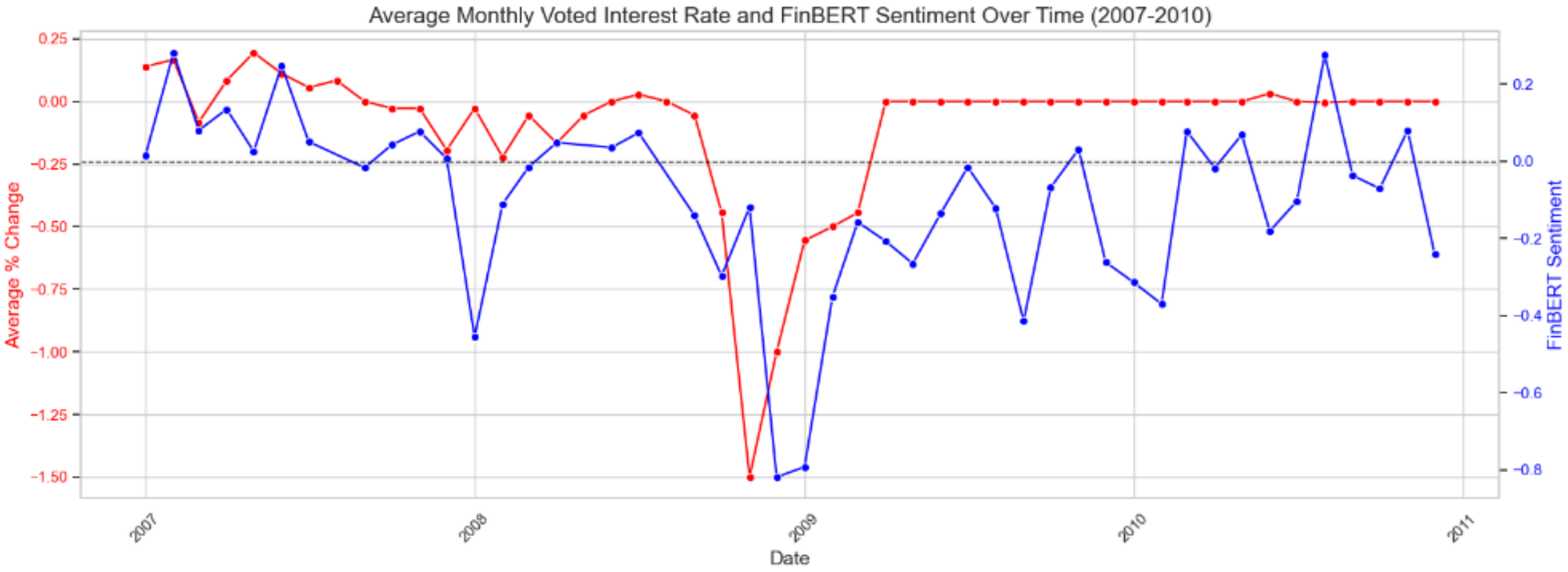
Average Monthly Sentiment Score (2018-2022)





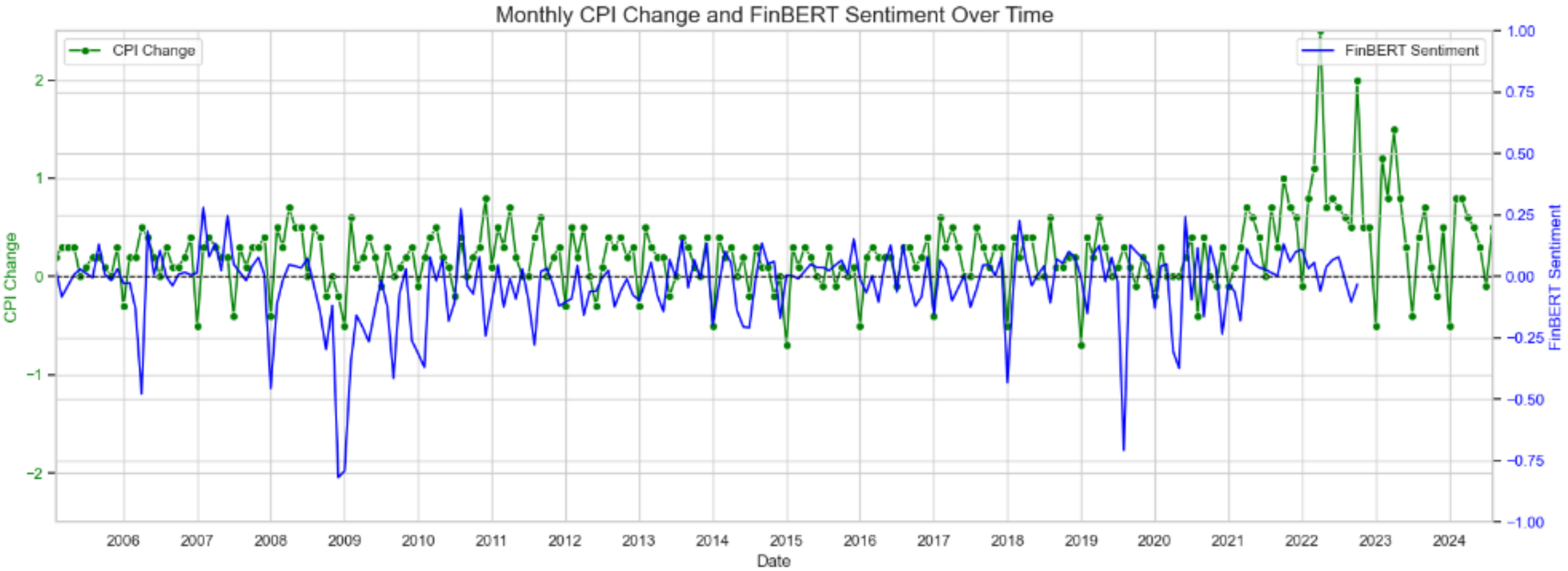
- The Pearson Correlation Coefficient is 0.25058, indicating fairly weak, positive correlation between sentiment scores and (average voted) interest rates.
- The associated P-value is 0.00006, which deems the observed correlation to be statistically significant.

Interest is set as a metric by the Bank of England to manage the economic activity, manage inflation and support economic stability. Therefore, we would not necessarily expect the sentiment of any given speech to significantly influence interest rates or vice versa, because it is more of an objective measure.



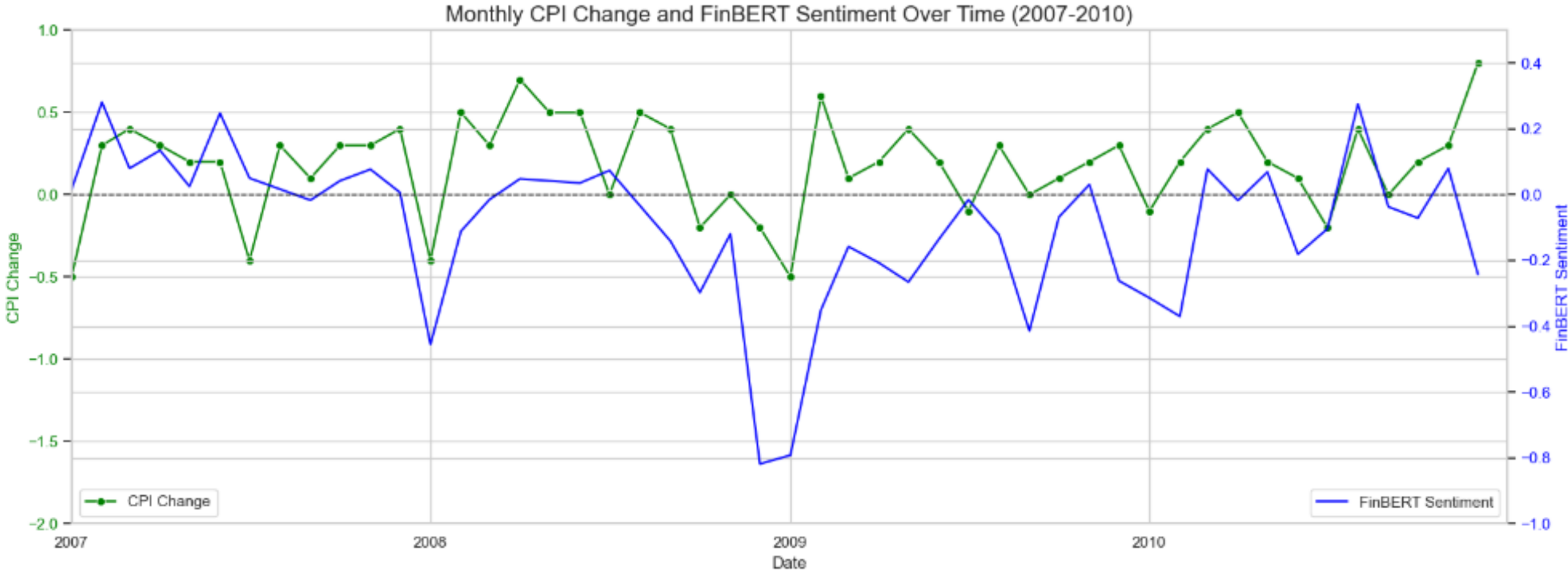
- The Pearson Correlation Coefficient is 0.42436, indicating stronger, positive correlation between sentiment scores and (average voted) interest rates, compared to the full timeframe.
- The associated P-value is 0.01237, which deems the observed correlation to be statistically significant.

Although typically Interest is mentioned in an objective manner, the speech sentiments and the interest changes show susceptibility to significant external economic events.



- The Pearson Correlation Coefficient is 0.1632, indicating weak, positive correlation between sentiment scores and interest rates.
- The associated P-value is 0.02, which deems the observed correlation to be statistically significant.

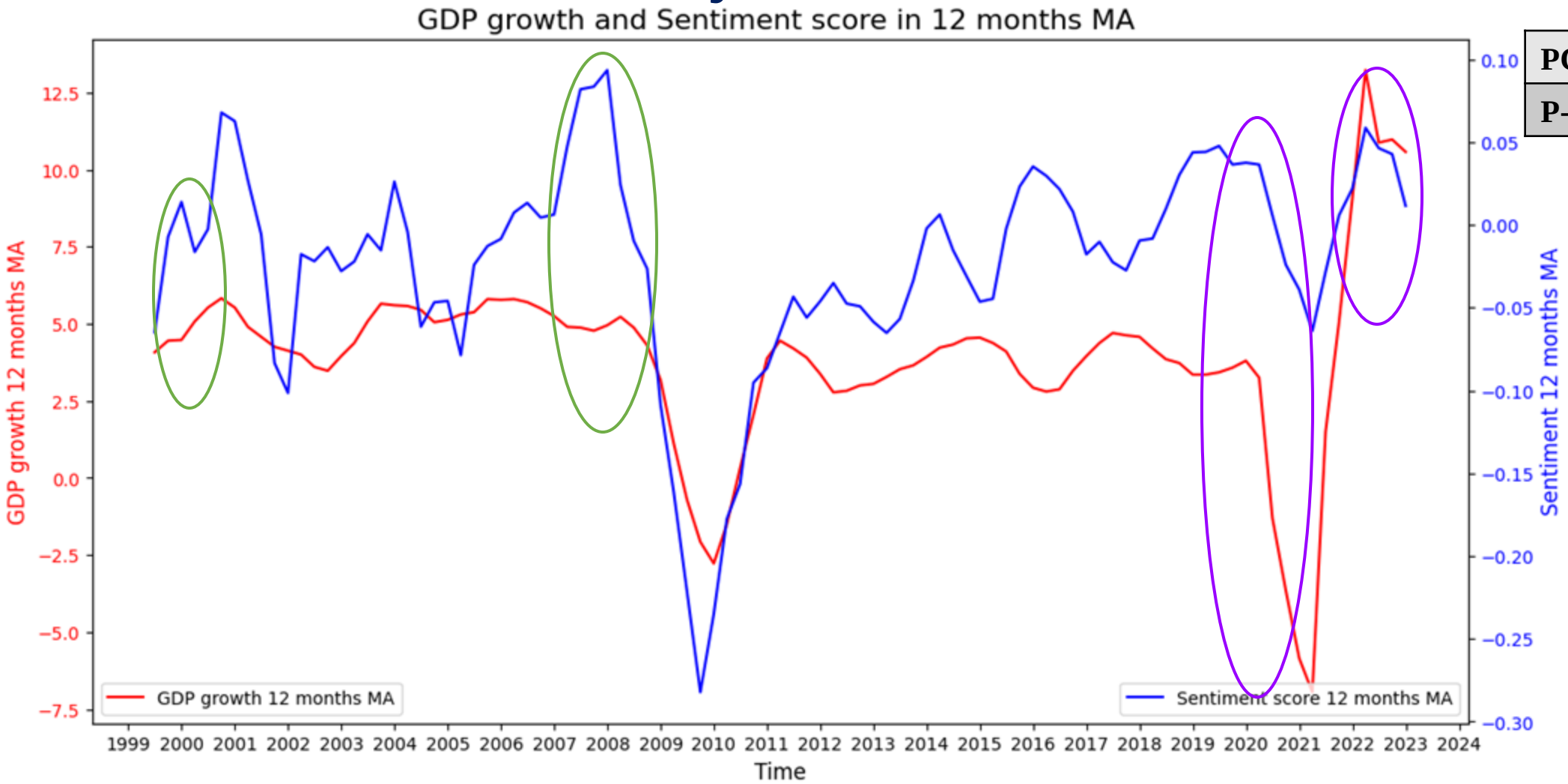
Inflation is a targeted metric, typically 2% p.a. Hence, we would not expect a significant correlation between the sentiment score and inflation rate change. Like interest, CPI lends itself to being more of an objective measure.



From 2007 through to 2010, the correlation observed between CPI and Sentiment score is far stronger, with the correlation coefficient measuring 0.4949. (P-value 0.0029)

This correlation, as was seen with interest rate and sentiment correlation, implies that speech sentiment and CPI changes are also affected by significant external economic events.

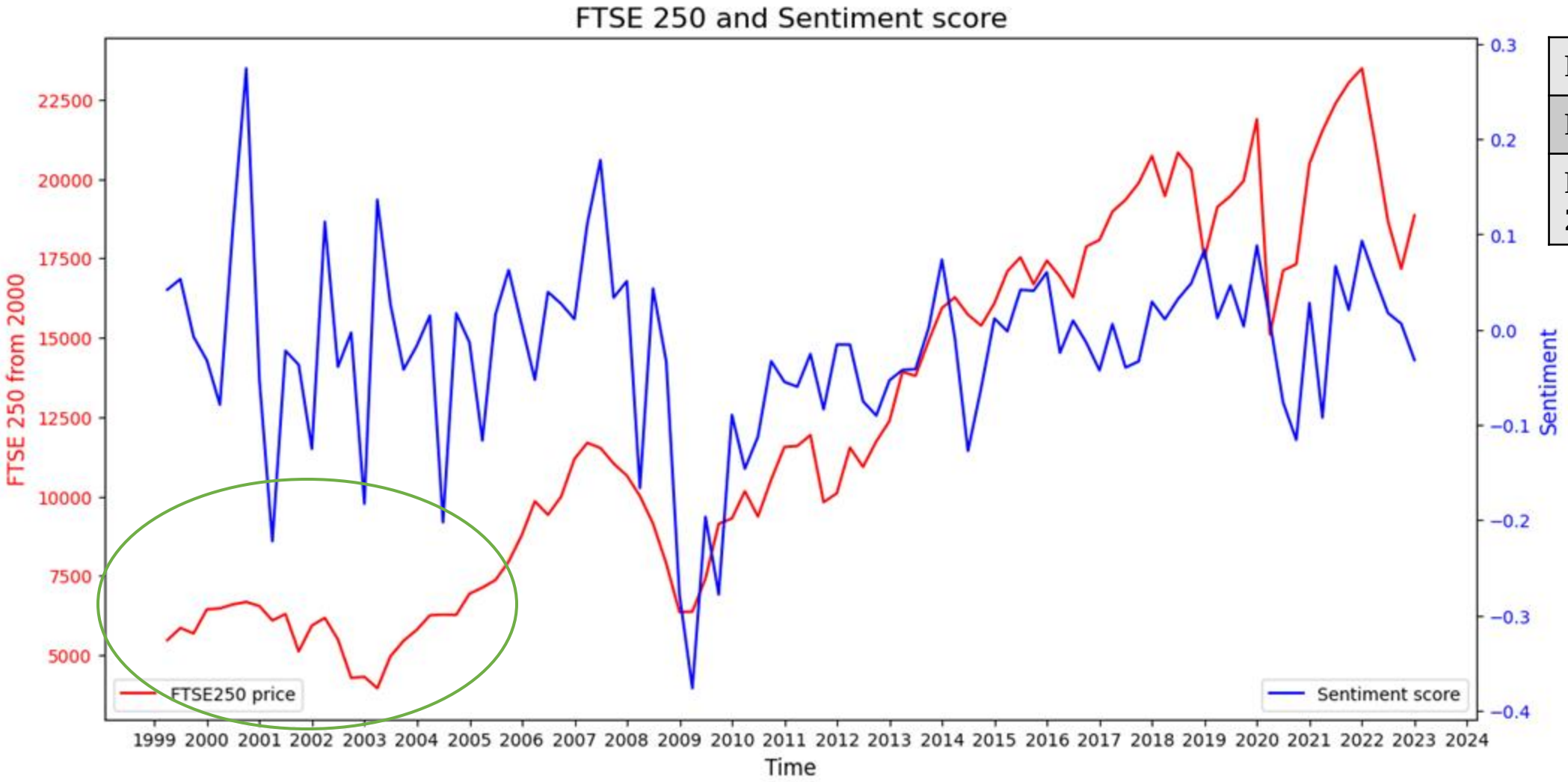
GDP & Sentiment Analysis



PCC (All)	0.526
P-value	4.27e-08

- GPD is a lagging indicator.
- Dotcom bubble and global recession sentiment dropped before GDP dropped
- Not applicable to Covid and Ukraine war possibly because non-financial event is not so predictable.

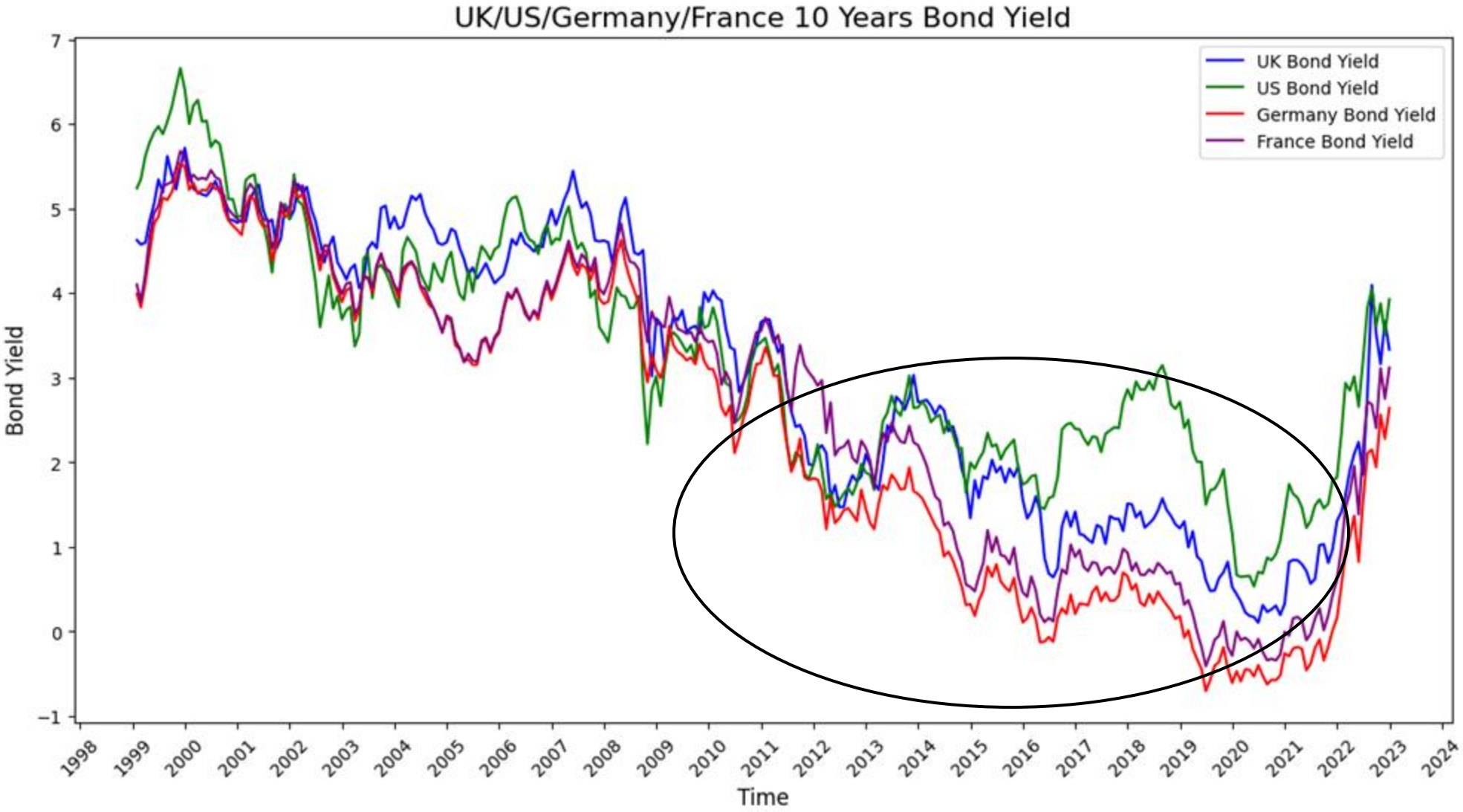
FTSE & Sentiment Analysis



PCC (All)	0.243
P-value	1.24e-05
PCC (2006-2023)	0.503

- FTSE is a leading indicator.
- Picked FTSE 250 instead of FTSE 100 even though FTSE 100 provides better correlation
- Rationale behind: Dotcom bubble had not much impact to FTSE 250.

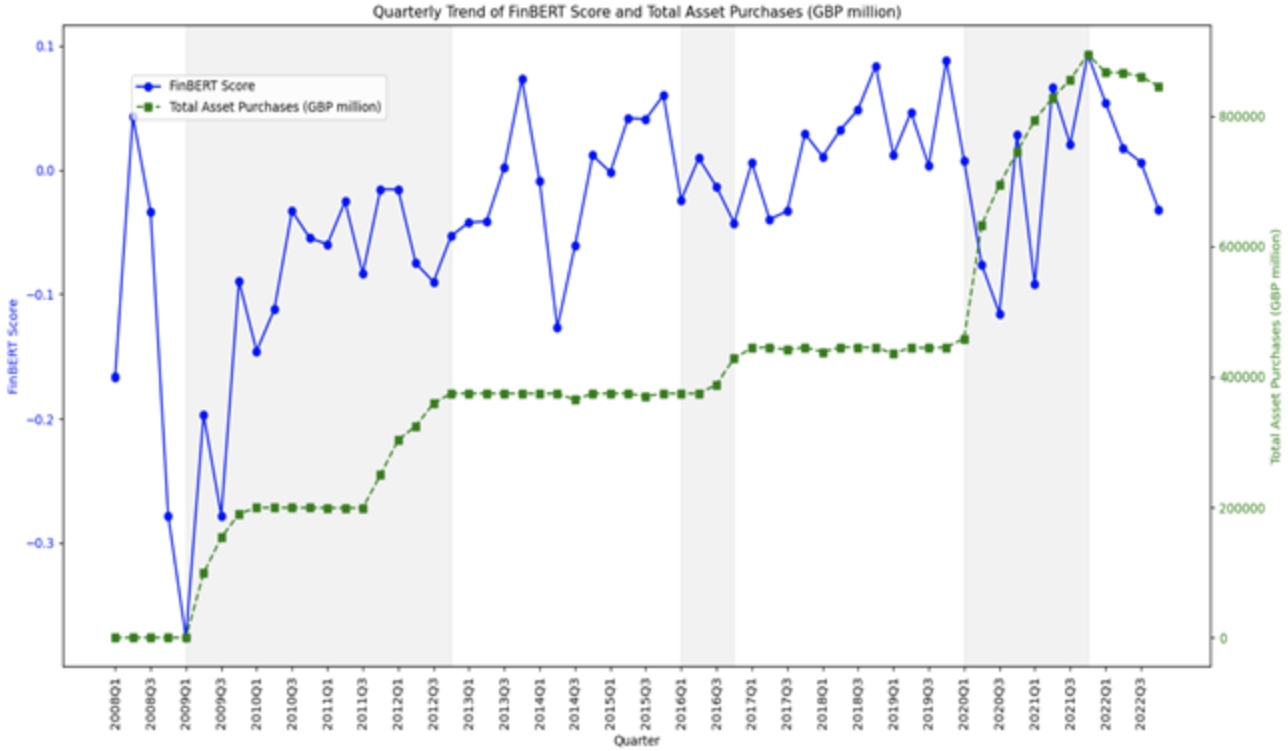
Bond Yield Comparison

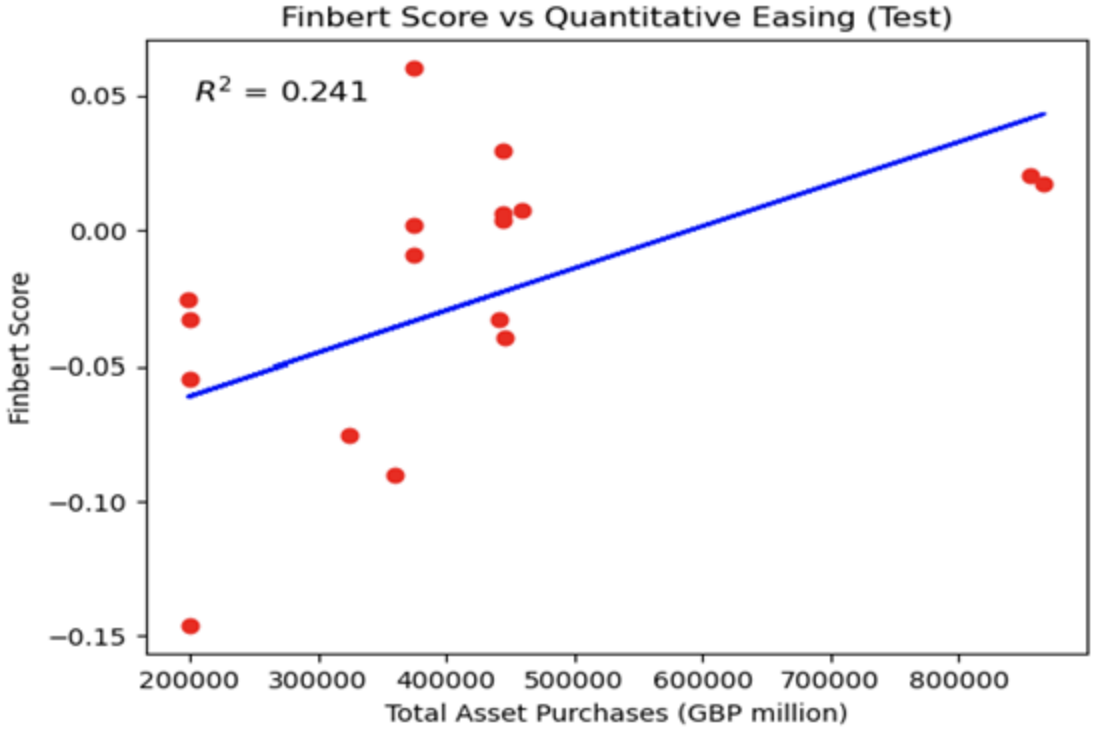


PCC (All)	-0.054
P-value	0.600

- All countries 10 years bond are on a similar trend.
- After global recession, all countries Interest Rate was lowered following the FED and a lot of QE policy was introduced.
- Data after global recession was sort of skewed due to the monetary policy.

FinBERT Score & Quantitative Easing





Global Financial Crisis - QE began in 2009 and continued till 2012.

Brexit - A new QE phase in 2016 including purchase of corporate bonds.

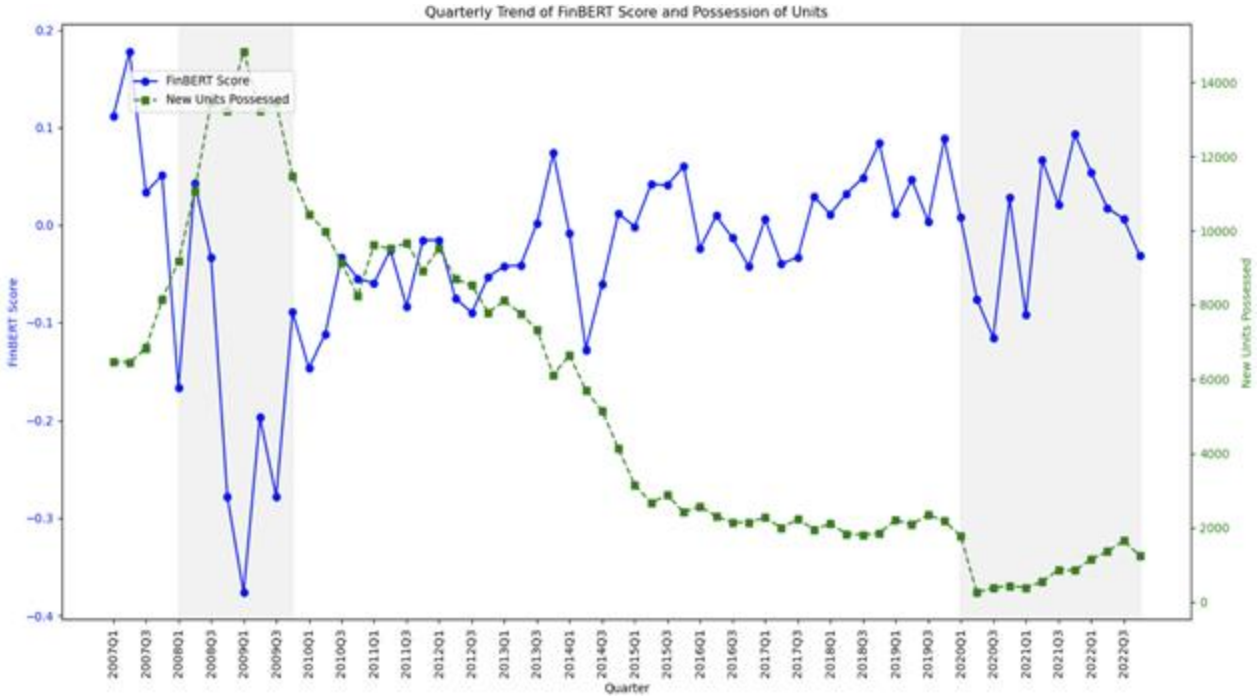
Covid Pandemic - QE5 launched in Mar'20 with GBP450bn added by Nov'21

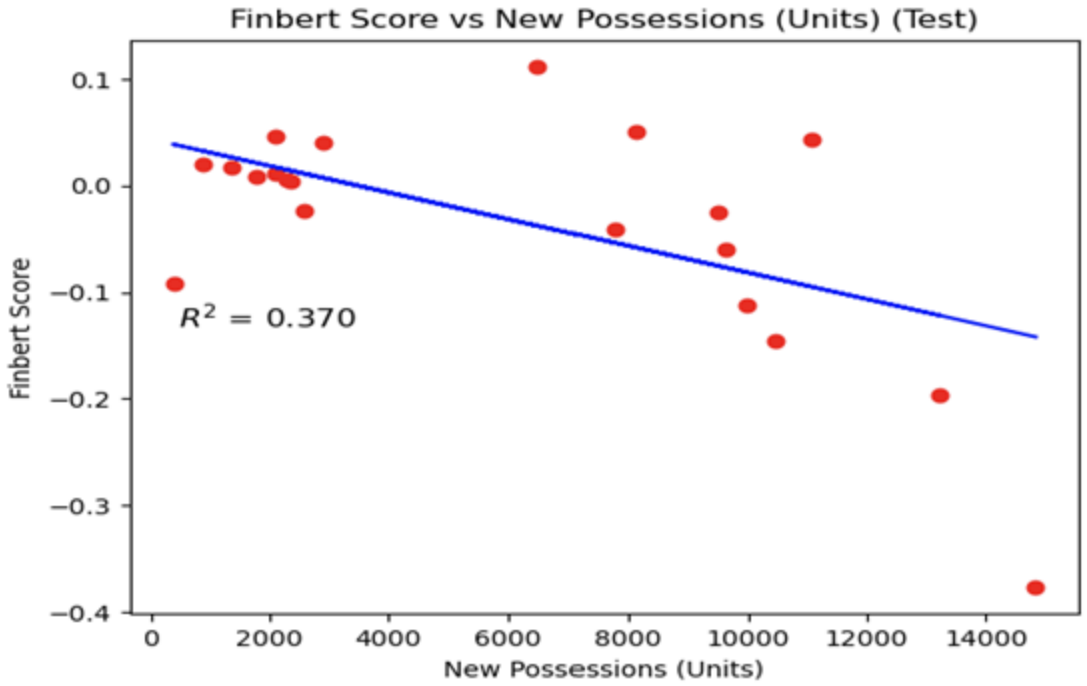
Pearson's correlation coefficient: 0.438 - Moderately positive correlation

P Value: 0.0008- Low P value indicates that the negative correlation is statistically significant

Global Financial Crisis - Correlation coefficient of 0.666 with p value of 0.10

FinBERT Score & Mortgage Defaults*





Global Financial Crisis (2008) - 100,000 properties repossessed.

Covid Pandemic (2020-2021) - Moratorium on repossessions.

Post Covid Pandemic - Rising interest rates and high inflation leading to rise in mortgage default rates.

Pearson's correlation coefficient: -0.587: Moderately negative correlation

P Value: 3.52e-07 - Low P value indicates the negative correlation is statistically significant.

Global Financial Crisis - Correlation coefficient of -0.811 with p value of 0.00014

Last Five Years - Correlation coefficient of 0.454 with p value of 0.0445

**https://www.fca.org.uk/data/mortgage-lending-statistics*

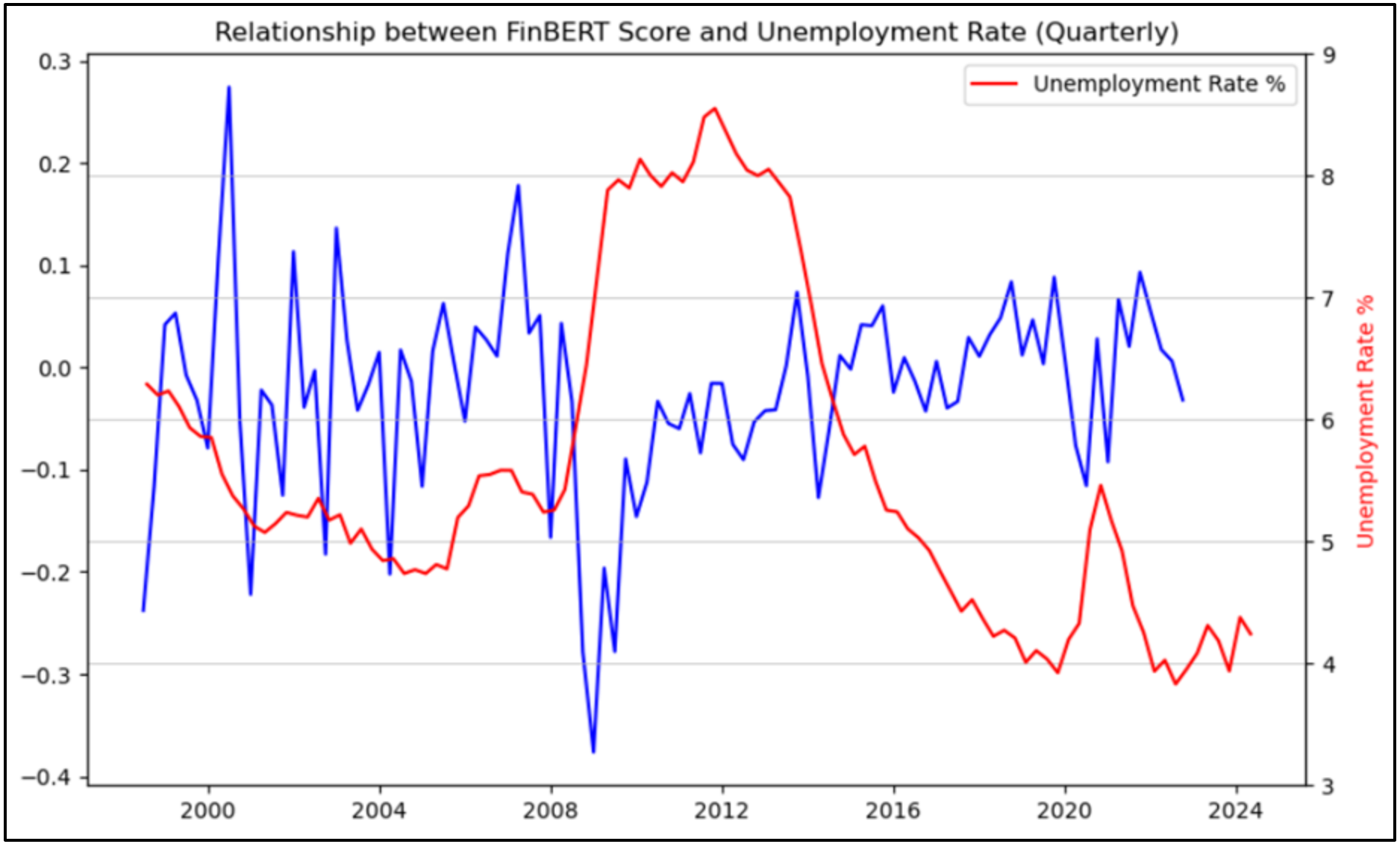
Sentiment & Unemployment Rate (1998 - 2022)

Correlation:

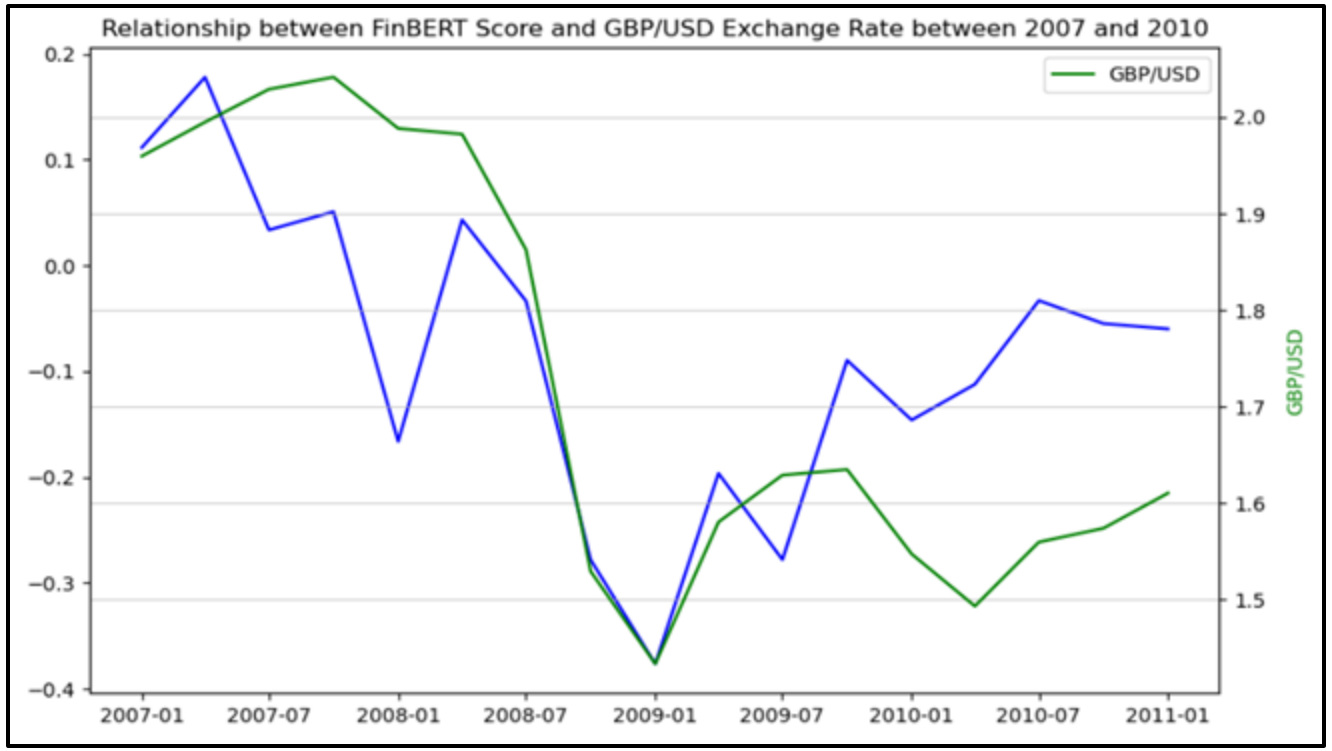
-0.357
Moderate negative

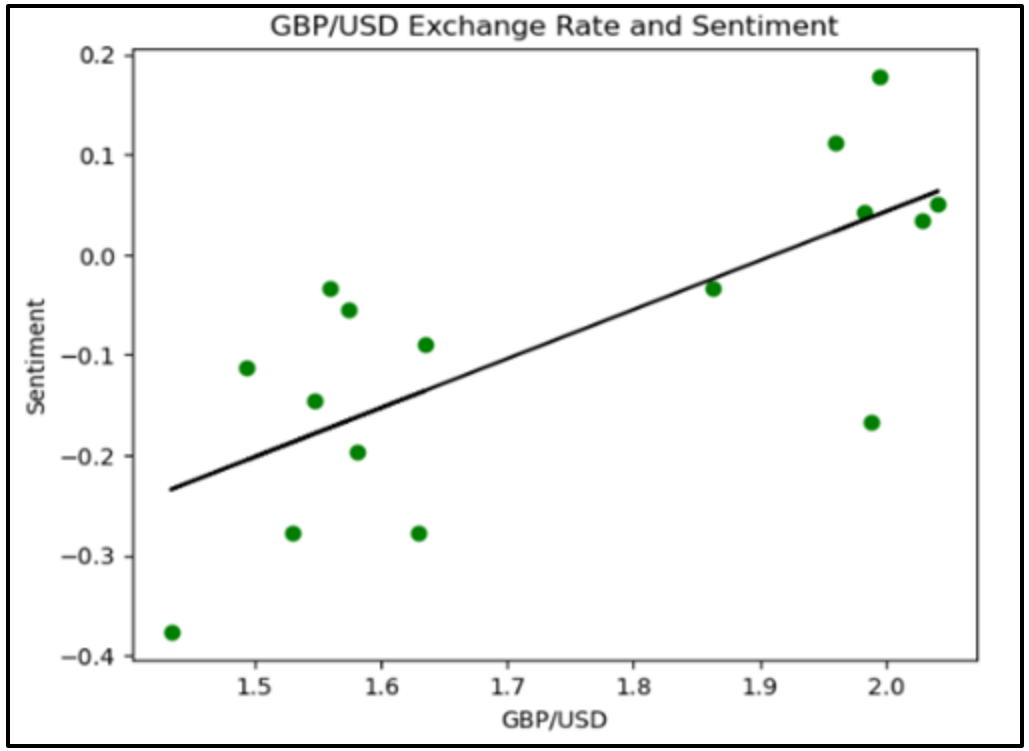
P-Value:

0.0003
Statistically Significant



Sentiment & GBP/USD Exchange Rate (2007 - 2010)





Correlation:

Correlation: **0.73** Strong positive
P-Value: **0.0013** Statistically Significant

Regression Results:

R-squared: **0.533**
Adj. R-squared: 0.500
P-value: **0.001** Statistically Significant

Correlation Table

Indicator Correlation with Speech Sentiment										
	Interest Rate Change	**CPI**	**GDP**	**FTSE 100**	**FTSE 250**	**10 Yr Bond Yield**	**QE (Total Asset Purchases)**	**Mortgage Defaults**	**Unemployment**	**GBP/USD Exchange Rate**
Correlation Coefficient	0.25058	0.1632	0.526	0.399	0.243	-0.054	0.438	-0.587	-0.357	-0.018
P-Value	0.0001	0.02	4.27E-08	5.77E-05	0.017	0.6	0.0008	3.52E-07	0.0003	0.8621
Indicator Correlation with Speech Sentiment: Financial Crisis (2007-2010)										
	Interest Rate	**CPI**	**GDP**	**FTSE 100**	**FTSE 250**	**10 Yr Bond Yield**	**QE (Total Asset Purchases)**	**Mortgage Defaults**	**Unemployment**	**GBP/USD Exchange Rate**
Correlation Coefficient	0.48423	0.3745	0.692	0.840	0.764	0.692	0.666	-0.811	-0.532	0.73
P-Value	0.0008	0.0113	2.14E-07	4.75E-05	0.001	0.003	0.102	0.00014	0.034	0.0013
Indicator Correlation with Speech Sentiment: 2018-2022										
	Interest Rate	**CPI**	**GDP**	**FTSE 100**	**FTSE 250**	**10 Yr Bond Yield**	**QE (Total Asset Purchases)**	**Mortgage Defaults**	**Unemployment**	**GBP/USD Exchange Rate**
Correlation Coefficient	0.04605	0.0786	0.692	0.556	0.420	0.088	-0.149	0.454	-0.373	0.153
P-Value	0.7867	0.5682	0.000176	0.0109	0.065	0.711	0.53	0.045	0.1049	0.51789

Conclusions & Recommendations

Conclusion	Recommendation
The correlation between speech sentiments and many economic metrics appears to be stronger following significant external economic events.	Establish a standardized dictionary to use in speeches during crisis periods. This consistent messaging can help mitigate unintended fluctuations in market behaviour triggered by speech sentiment.
There are inconsistencies with time frames for which different economic metrics are available.	Continue to release clear information on economic indicators in a clear time period to increase transparency and maintain data availability.
The sentiment scores observed are typically neutral which avoids extreme sentiment. This is in line with the low predictive power as highlighted in the regression modelling.	Conduct scenario-based analysis to make predictions, which considers a range of future economic metric changes and their potential impact.

Future Analysis

- Due to limited data beyond 2022, the impact of the Ukraine – Russia conflict has not yet been analysed.
- Analysing the impact of different speakers within the Bank of England could yield valuable insights.
- Extend analysis to explore correlations between overseas bank speech sentiment scores and their domestic economic metrics.

Thank
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